

AMFU-Net Replication

No Dataset, No Fusion — How the Replacement Preprocessing Works

Final Year Project — Supervisor Briefing

Why There's No Dataset, in Brief

- AMFU-Net needs paired reflection + transmission USCT images with lesion masks — searched across Springer, IEEE, Elsevier, PubMed, ACM, Nature, and Wiley, and it does not exist publicly
- Reason: the hardware that captures both signals (ring-array USCT) exists at only a handful of research sites, and every one of them keeps patient data inside IRB-controlled clinical trials
- The AMFU-Net authors state directly that they do not have permission to share their data
- Closest public dataset, OpenBreastUS (2025), is synthetic wave-simulation data with no lesion masks — not usable for segmentation
- Conclusion: the fusion step cannot be replicated. The rest of this briefing is about what replaces it, and why that replacement is a defensible choice, not a downgrade

The Replacement: Single-Image Enhancement

- Dataset moves to public single-modality sets — BUSI, BUS-UCLM, BUS-BRA
- AMFU-Net's DAG attention gates and MFEF module stay in the architecture unchanged
- What's new: a preprocessing stage — speckle reduction, then CLAHE — takes over the role the second (transmission) image used to play in the original paper
- This is not a new idea invented for this project — it's a well-established preprocessing pattern in breast ultrasound segmentation literature, traced back to a specific 2018 paper (slide 9)
- The rest of this briefing explains exactly how each part of that preprocessing stage works, and how to replicate it

How CLAHE Works (Contrast Limited Adaptive Histogram Equalization)

- Plain histogram equalization stretches contrast globally across the whole image — one transformation applied everywhere, regardless of local content
- Adaptive Histogram Equalization (AHE) instead divides the image into small tiles and equalizes each tile's histogram separately, giving contrast that responds to local structure
- Problem: in a flat, low-variance tile, AHE's transformation curve becomes very steep, so it massively amplifies whatever small variations are there — including speckle noise
- CLAHE's fix: before building each tile's histogram, it clips any bin taller than a fixed clip limit, and spreads the clipped-off pixel count evenly across the rest of the histogram — this caps how steep the local transform can get
- Tiles are equalized independently and then bilinear interpolation blends their borders, so there's no visible blocking between tiles
- Net effect: local contrast around lesion boundaries and tissue texture improves, without the noise blow-up that plain AHE causes

Why Speckle Needs Its Own Filter

- Speckle is not sensor noise — it's a granular interference pattern from coherent sound waves scattering off tissue and interfering with each other, so it's multiplicative, not additive
- It sits directly on top of real tissue texture rather than being a separate layer, which is what makes it hard to remove cleanly
- Generic denoisers don't handle this well: a Gaussian blur or plain median filter either leaves speckle behind, or blurs away real lesion edges along with it, because they can't distinguish speckle statistics from a genuine boundary
- This is why ultrasound preprocessing pipelines use filters purpose-built for speckle's statistics — SRAD and DCAIP, covered next — instead of a generic image denoiser

SRAD — How It Works

- Speckle Reducing Anisotropic Diffusion, Yu & Acton (2002) — this is the exact filter used in the paper this whole approach is replicating (slide 9)
- Built on anisotropic diffusion: the image is smoothed by solving a diffusion equation over several iterations, similar to how heat spreads through a material
- At every pixel, SRAD computes the instantaneous coefficient of variation (ICOV) — a local statistic built from the gradient and Laplacian, derived specifically for multiplicative speckle rather than generic additive noise
- Where ICOV is low (flat, speckled tissue) the diffusion coefficient is pushed close to 1 → strong smoothing is applied
- Where ICOV is high (a real boundary, like a lesion edge) the diffusion coefficient drops toward 0 → smoothing is halted there, so the edge is preserved rather than blurred away
- Run iteratively — more iterations remove more speckle, while genuine edges stay locked in place throughout the process

DCAIP — A Modern Alternative

- Deep Content-Aware Image Prior (DCAIP) — from Lee et al. (2022), "Speckle Reduction via Deep Content-Aware Image Prior for Precise Breast Tumor Segmentation in an Ultrasound Image," IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control — a despeckling method aimed directly at breast tumour segmentation, useful because there are no clean/noisy ultrasound image pairs available to train a supervised denoiser
- Built on the deep image prior idea: a randomly initialised CNN is fit directly to one single noisy image — no training dataset is required at all
- A CNN naturally learns structured content faster than it learns to reproduce random noise during optimisation, so stopping the fitting early — before the network starts reproducing the noise — yields a denoised image
- DCAIP adds a content-aware attention module with super-pixel pooling: the image is grouped into small, perceptually uniform super-pixel regions, and the network's attention is guided by these regions so flat areas get smoothed more aggressively while boundaries are respected
- Trade-off versus SRAD: no hand-tuned diffusion parameters and better handling of irregular lesion shapes, at the cost of fitting a small CNN per image rather than running a fast iterative filter

SRAD vs DCAIP — Side by Side

- Speed — SRAD: fast, a few iterations of a simple diffusion filter. DCAIP: slower, fits a small CNN separately for every image
- Parameter tuning — SRAD: needs hand-set iteration count and step size. DCAIP: no diffusion parameters to tune, but network/training settings still matter
- Compute cost — SRAD: negligible, runs on CPU easily. DCAIP: needs a GPU pass per image to be practical at dataset scale
- Handling irregular lesion shapes — SRAD: works well on typical rounded lesions, can under-preserve unusual boundary shapes. DCAIP: adapts better to irregular shapes via superpixel-guided attention
- Evidence base for this exact use case — SRAD: proven directly — it's the filter Almajalid et al. (2018) used in the paper this project is replicating. DCAIP: also proven directly on breast tumour segmentation — Lee et al. (2022) built and tested it specifically for that task, not just general despeckling
- Recommendation — start with SRAD as the primary, proven route; add DCAIP later as a second ablation row if time allows, to show more than one despeckling strategy was tested

Pipeline Order: Despeckle First, Then CLAHE

- Order matters, and it is consistent across the papers that report real gains from combining the two
- Despeckle first (SRAD or DCAIP) — removing speckle necessarily softens the image slightly, as a side effect of the smoothing itself
- CLAHE second — the clipped local-contrast step then restores and sharpens texture and boundary detail that the despeckling step softened
- Doing it in reverse — CLAHE before despeckling — enhances local contrast on a still-noisy image, which stretches the speckle grain along with real structure, making it harder to remove cleanly afterward

Where This Worked: Almajalid et al. (2018)

- Paper: "Development of a Deep-Learning-Based Method for Breast Ultrasound Image Segmentation," IEEE ICMLA 2018 (DOI: 10.1109/ICMLA.2018.00179)
- What they did: applied contrast enhancement (histogram equalization) plus SRAD-based speckle reduction to raw BUS images before training — same two-stage idea covered on the last three slides
- Fed the preprocessed images into a standard U-Net for lesion segmentation, then filtered false-positive regions under the assumption of a single tumour per image
- Trained and validated on a combined dataset of 993 breast ultrasound images drawn from three public collections
- This is the paper the entire "contrast enhancement + speckle reduction" preprocessing lineage in later BUS segmentation papers traces back to — it is cited as the origin point in nearly every architecture paper that follows it

Replication Plan for This Project

- Step 1 — Implement SRAD on raw BUSI / BUS-UCLM images, run for a fixed number of iterations (start with the iteration count used in Yu & Acton, then tune)
- Step 2 — Apply CLAHE on top of the despeckled output; tile size and clip limit are the two tunable parameters, start from values used in the ultrasound-specific papers above
- Step 3 — Feed the preprocessed images into the existing DAG + MFEF architecture, unchanged, exactly as the raw-image baseline would be fed
- Step 4 — Run a controlled ablation: same architecture, same data split, one run on raw images and one run on preprocessed images — report the Dice/IoU difference, which is the exact comparison Almajalid et al. and the later papers used to justify this preprocessing step
- Step 5 — If time allows, swap SRAD for DCAIP as a second ablation row, to show more than one despeckling strategy was tested

References

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- SegNet + CLAHE / Gaussian blur / Gabor filtering preprocessing pipeline on BUSI — SpringerLink
- Enhancement of Ultrasound Images with Bilateral Filter and Rayleigh CLAHE (CLAHE-RD) — ResearchGate
- Feature Enhancement in Medical Ultrasound Videos Using Contrast-Limited Adaptive Histogram Equalization — PMC
- AMFU-Net paper (source study) — states authors do not have permission to share their dataset
- OpenBreastUS: Benchmarking Neural Operators for Wave Imaging Using Breast USCT — arXiv, 2025